

Experiment 5

Aim: To apply explainable AI (XAI) methods (SHAP & LIME) for interpreting model predictions and evaluate fairness using Fairlearn.

Objective

- Generate **global** (SHAP feature importance) and **local** (LIME) explanations of model predictions.
- Audit bias in ML models across sensitive features (e.g., gender, race).
- Propose fairness mitigation strategies if bias is detected.

Detailed Steps

1. Dataset & Model Selection

- Use a dataset with potential fairness aspects
- Train a classifier/regressor.

2. Explainability with SHAP

- Install and apply SHAP (shap.TreeExplainer or KernelExplainer).
- Generate **global explanations**: feature importance summary plots, dependence plots.
- Interpret which features most influence predictions.

3. Explainability with LIME

- Install and run LIME (lime.lime_tabular).
- Generate **local explanations** for individual predictions.
- Visualize contribution of features for specific samples.

4. Fairness Audit with Fairlearn

- Define sensitive attribute(s) (e.g., gender, race).
- Use **Fairlearn's metrics** (demographic parity difference, equalized odds difference).
- Generate a fairness report comparing performance across groups.

5. Bias Mitigation

- If bias detected, propose strategies such as:
 - Pre-processing (reweighting data, sampling).
 - In-processing (fairness constraints in training).
 - Post-processing (threshold adjustments).

Open-Source Tools

- **SHAP** – Global feature importance.
- **LIME** – Local explanation of predictions.
- **Fairlearn** – Fairness metrics and bias mitigation.

Deliverables

- SHAP plots (summary, dependence).
- LIME local explanation visualizations.
- Fairness audit report (metrics, visualizations).

Conclusion